MTH 200 **Chapter 5: Integration**

5.4 Applications of Integration

Theorem (Net Change Theorem): The integral of a rate of change is the *net change*:

$$\int_a^b F'(x)\,dx = F(b) - F(a)$$

Examples:

- Integral of *volume/sec* is net change in *volume*.
- Integral of *tons of pollution/hour* is net change in *tons of pollution*.
- Integral of *velocity* is the net change in *position*.

Example 1: Suppose that oil starts leaking from a tank at a rate of $100-5t$ gallons per minute at time *t* from when the leaking starts.

a) What does $\int_0^{120} (100-5t)\,dt$ represent?

This integral represents the total number of gallons leaked from the tank within the first 2 hours (120 minutes).

b) Find the total number of gallons leaked in the first half-hour.

$$\int_0^{30} (100-5t)\,dt = \left(100t + \frac{5}{2}t^2\right)\Big|_0^{30} = 100(30) - \frac{5}{2}(30)^2 - 0 = 3000 - 2250 = 750\,gallons$$

Example 2: At 4:00am, the layer of ice on a lake was 6in. thick. Its thickness is increasing at a rate of $0.2t$ inches per hour where *t* represents the number of hours since 4:00am.

a) How thick was the ice at noon that same day?

Thickness at noon = current thickness + net change over 8 hours

$$\text{Thickness at noon} = 6 + \int_0^8 0.2t\,dt = 6 + \left(0.1t^2\right)\Big|_0^8 = 6 + 0.1(8)^2 - 0.1(0)^2 = 6 + 6.4 = 12.4\,inches$$

b) How much thickness was gained from 6:00am to noon that same day?

$$Thickness\ gained = \int_2^8 0.2t\,dt = \left(0.1t^2\right)\Big|_2^8 = 0.1(8)^2 - 0.1(2)^2 = 6.4 - 0.4 = 6\,inches$$

Note: Finding the *total distance* travelled is different than computing the *total displacement*.

- *Displacement* represents a *Net Change* and so quantities of distance are measured as positive or negative.
- With *total distance*, all quantities are regarded as positive.

Example 3:

A particle moves along a line so that its velocity at time t is $v(t)=t^2-t-6$ (measured in meters per second).

a) Find the displacement of the particle during the time period $1 \le t \le 4$.

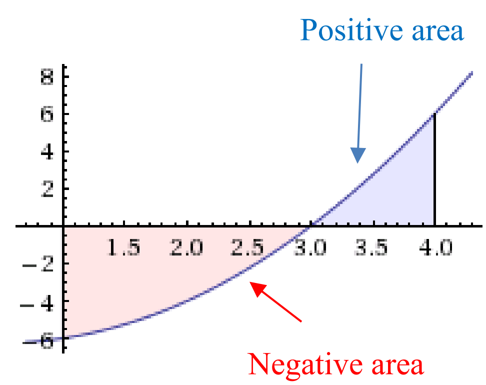
$$displacement = \int_1^4 \left(t^2-t+6\right)dt = \left(\frac{t^3}{3}-\frac{t^2}{2}+6t\right)\Bigg|_1^4 = \left[\frac{4^3}{3}-\frac{4^2}{2}+6(4)\right]-\left[\frac{1^3}{3}-\frac{1^2}{2}+6(1)\right]$$

$$=\frac{64}{3}-8+24-\frac{1}{3}+\frac{1}{2}+6$$

$$=-\frac{9}{2}\,meters$$

The total *net change* in the particle's position is $-\frac{9}{2}\,meters$. This means that the particle moved 4.5*m* to the left.

b) Find the distance traveled during this time period.



We must first find when the particle is at rest.

$$v(t)=t^2-t-6=(t-3)(t-2)$$

$v(t)=0$ only when $t=3$.

We can see that when $t<3$, $v(t)<0$ so we can simply multiply this portion of the curve by (-1) to make it positive.

$$\text{Total Distance Travelled } = \int_1^4 |v(t)|\,dt = \int_1^3 -v(t)\,dt + \int_3^4 v(t)\,dt = \int_1^3 \left(-t^2+t+6\right)dt + \int_3^4 \left(t^2-t-6\right)dt$$

$$=\left(-\frac{t^3}{3}+\frac{t^2}{2}+6t\right)\Bigg|_1^3+\left(\frac{t^3}{3}-\frac{t^2}{2}-6t\right)\Bigg|_3^3$$

$$=\left[-9+\frac{9}{2}+18\right]-\left[-\frac{1}{3}+\frac{1}{2}+6\right]+\left[\frac{64}{3}-8-24\right]-\left[9-\frac{9}{2}-18\right]$$

$$=\frac{61}{6}\,meters$$